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Homework 10
Dr. Brett Kotschwar - Fall 2023

Exercise 6.2: Suppose $f \geq 0$, f is continuous on $[a,b]$, and $\int_a^b f(x)dx = 0$.
Claim: $f(x) = 0$ for all $x \in [a,b]$.
Proof: By the Fundamental Theorem of Calculus I, let $g : [a,b] \to \mathbb{R}$ be such that $g' = f$. Then, $\int_a^b f(t)dt = g(b) - g(a) = 0$. Thus, $g(b) = g(a)$. But, $f = g' \geq 0$, so g is monotone increasing on $[a,b]$. Hence g is constant on $[a,b]$. Since g is constant on $[a,b]$, if $x \in (a,b)$, then $g'(x) = f(x) = 0$, as

$$g'(x) = \lim_{t \to x^+} \frac{g(t) - g(x)}{t - x} = \lim_{t \to x^-} \frac{g(t) - g(x)}{t - x} = \frac{0}{t - x} = 0.$$

Further, if $x = a$, then

$$g'(a) = \lim_{t \to a^+} \frac{g(t) - g(a)}{z - a} = \lim_{t \to a} \frac{g(t) - g(a)}{t - a} = 0.$$

And, likewise, if $x = b$, then

$$g'(b) = \lim_{t \to b^-} \frac{g(t) - g(b)}{t - b} = \lim_{t \to a} \frac{g(t) - g(b)}{t - b}.$$

Thus, it follows that $f(x) = 0$ for all $x \in [a,b]$. ■

Exercise 6.8: Suppose $f \in \mathscr{R}$ on $[a,b]$ for every $b > a$ where a is fixed. Define

$$\int_a^\infty f(x)dx = \lim_{b \to \infty} \int_a^b f(x)dx$$

if this limit exists (and is finite). Assume that $f(x) \geq 0$ and that f decreases monotonically on $[1,\infty)$.
Claim: $\int_1^\infty f(x)dx$ converges if and only if $\sum_{n=1}^\infty f(n)$.
Proof: First, suppose $\int_1^\infty f(x)dx$ converges. Let $\int_1^\infty f(x)dx = \lim_{b\to\infty} \int_1^b f(x)dx = C$. Since f is decreasing for all $x \in [1,\infty)$, by an integral property (Lecture 21), for any $x \in [n-1, n]$, where $n \in \mathbb{N}$, $f(n) \leq f(x) \leq f(n-1)$. Thus, $f(n) = f(n)(n - (n-1)) \leq \int_{n-1}^n f(x)dx \leq f(n-1)(n-(n-1)) = f(n-1)$. Now, let $\{S_n\}$ be a sequence of partial sums, where $S_n = \sum_{k=1}^n f(k)$. Then, by the previous inequality and an integral property (Lecture 21), $S_n = \sum_{k=1}^n f(k) = f(1) + \sum_{k=2}^n f(k) \leq f(1) + \sum_{k=2}^n \int_{k-1}^k f(x)dx = f(1) + \int_1^2 f(x)dx + \int_2^3 f(x)dx + \ldots + \int_{n-1}^n f(x)dx = f(1) + \int_1^n f(x)dx < f(1) + \int_1^\infty f(x)dx = f(1) + C,$ for all $n \in \mathbb{N}$.Hence $\{S_n\}$ is bounded above. Since $f(x) \geq 0$, $\{S_n\}$ is monotonically increasing. Therefore, $\{S_n\}$, or $\sum_{n=1}^\infty f(n)$, converges.

Conversely, suppose for contrapositive that $\int_1^\infty f(x)dx$ is not finite, i.e. $\int_1^n f(x)dx$ diverges. Then, $\int_1^n f(x)dx$ diverges to $+\infty$ as $n \to \infty$ because $f(x) \geq 0$. So, $S_{n-1} = \sum_{k=2}^{n-1} f(k-1)$, and, recalling our earlier inequality, $S_{n-1} = \sum_{k=2}^{n-1} f(k-1) \geq \sum_{k=2}^{n-1} \int_{k-1}^k f(x)dx =$

$\int_1^{n-1} f(x)dx$. Since $\int_1^n f(x)dx$ diverges as $n \to \infty$, $\int_1^{n-1} f^{-1}(x)dx \to \infty$ as $n \to \infty$. So, by comparison theorem, it follows from the previous inequality that $\{S_{n-1}\}$ and thus $\{S_n\} = \sum_{n=1}^{\infty} f(n)$ diverges. ■

Exercise 6.10: Let p and q be positive real numbers such that

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Claim (a): If $u \geq 0$ and $v \geq 0$, then $uv \leq \frac{u^p}{p} + \frac{v^q}{q}$, and equality holds if and only if $u^p = v^q$.
Proof: If $u = 0$ or $v = 0$, then showing the inequality and condition for equality is trivial: if $u = 0$, then $0 \leq \frac{0}{p} + \frac{v^q}{q} = \frac{v^q}{q}$, and $\frac{v^q}{q} = 0$ is true if and only if $v^q = 0 = 0^p = u^p$.

Now, suppose $v > 0, u > 0$. Fix $v > 0$, and let $f(u) = \frac{u^p}{p} + \frac{v^q}{q} - uv$. We can show that $f(u) \geq 0$. $f'(u) = u^{p-1} - v = 0$ if $u = v^{\frac{1}{p-1}}$. So, by the extreme value theorem, $u = v^{\frac{1}{p-1}}$ is a critical point of $f(u)$. Also, since $f''(u) > 0$, $u = v^{1/(p-1)}$ is a minimum. And, $f(u) \geq f\left(v^{\frac{1}{p-1}}\right) = \frac{v^{\frac{p}{p-1}}}{p} + \frac{v^q}{q} - v^{1+\frac{1}{p-1}} = \left(\frac{1}{p} + \frac{1}{q} - 1\right) u^p - 0$. Thus, $u^p = v^{\frac{p}{1-p}} = v^{1-\frac{1}{p}} = v^q$. ■

Claim (b): If $f \in \mathscr{R}(\alpha), g \in \mathscr{R}(\alpha), f \geq 0, g \geq 0$, and

$$\int_a^b f^p d\alpha = 1 = \int_a^b g^q d\alpha,$$

then $\int_a^b fgd\alpha \leq 1$.
Proof: b) Since $fg \leq \frac{f^p}{p} + \frac{g^q}{q}$, by part (2) of the properties of integrals theorem (Lecture 21),

$$\int_a^b fgd\alpha \leq \int_a^b \frac{f^p}{p} + \frac{q^q}{q} d\alpha = \frac{1}{p}\int_a^b f^p d\alpha + \frac{1}{q}\int_a^b g^q d\alpha = \frac{1}{p} \cdot 1 + \frac{1}{q} \cdot 1 = \frac{1}{p} + \frac{1}{q} = 1. \blacksquare$$

Claim (c): If f and g are complex functions in $\mathscr{R}(\alpha)$, then

$$\left|\int_a^b fgd\alpha\right| \leq \left\{\int_a^b |f|^p d\alpha\right\}^{1/p} \left\{\int_a^b |g|^a d\alpha\right\}^{1/a}.$$

Proof: Let f and g be complex functions in $\mathscr{R}(\alpha)$. By Theorem 6.13(b), $fg, |f|, |g|, |fg| \in \mathscr{R}(\alpha)$ and $\left|\int_a^b fgd\alpha\right| \leq \int_a^b |fg|d\alpha \leq \int_a^b |f||g|d\alpha$. We can show that

$$\int_a^b |f||g|d\alpha \leq \left(\int_a^b |f|^p d\alpha\right)^{1/p} \left(\int_a^b |g|^q d\alpha\right)^{1/g}.$$

The inequality holds if

$$\int_a^b \frac{|f|}{\left(\int_a^b |f|^p d\alpha\right)^{1/p}} \frac{|g|}{\left(\int_a^b |g|^q d\alpha\right)^{1/g}} d\alpha \leq 1.$$

Let $f^* = \frac{|f|}{\left(\int_a^b |f|^p d\alpha\right)^{1/p}}$ and $g^* = \frac{|g|}{\left(\int_a^b |g|^b d\alpha\right)^{1/g}}$. Note that $f^*, g^* \geq 0$. By part B, the previous inequality is true if $\int_a^b (f^*)^p \, d\alpha = \int_a^b (g^*)^p \, d\alpha = 1$. Observe that

$$\int_a^b (f^*)^p \, d\alpha = \int_a^b \left(\frac{|f|}{\left(\int_a^b |f|^p d\alpha\right)^{1/p}} \right)^p d\alpha = \frac{1}{\left(\int_a^b |f|^p d\alpha\right)} \int_a^b |f|^p d\alpha = 1.$$

Repeating those steps for g^*, we also find that $\int_a^b (g^*)^p \, d\alpha = 1$. Thus, our condition is satisfied.

$$\text{Therefore, } \int_a^b |f||g| d\alpha \leq \left(\int_a^b |f|^p d\alpha \right)^{1/p} \left(\int_a^b |g|^p d\alpha \right)^{1/q}. \blacksquare$$

Exercise 6.11: Let α be a fixed increasing function on $[a, b]$. For $u \in \mathscr{R}(\alpha)$, define

$$\|u\|_2 = \left\{ \int_a^b |u|^2 d\alpha \right\}^{1/2}.$$

Claim: Suppose $f, g, h \in \mathscr{R}(\alpha)$. Then, $\|f - h\|_2 \leq \|f - g\|_2 + \|g - h\|_2$.
Proof: By theorem 1.37, $|f - h| \leq |f - g| + |g - h|$. Squaring $\|f - h\|_2$, we get

$$\left(\left(\int_b^b |f - h|^2 d\alpha \right)^{1/2} \right)^2 = \int_a^b |f - h|^2 d\alpha = \int_{ba}^b \big((f - g) + (g - h)^2 d\alpha$$
$$= \int_a^b |f - g + g - h|^2 d\alpha = \int_a^b (f - g)^2 + 2(f - g)(g - h) + (g - h)^2 d\alpha$$

$$\leq \int_a^b |f - g|^2 + 2|f - g\|g - h| + |g - h|^2 d\alpha = \|f - g\|_2^2 + 2 \int_a^b |f - g||g - h| da + \|g - h\|^2.$$

So, by Theorem 6.10(c), we have

$$\|f - g\|_2{}^2 + 2 \int_a^b \big| f - g\|g - h \mid da + \|g - h\|_2{}^2$$
$$\leq \|f - g\|_2{}^2 + 2 \left\{ \int_a^b |f - g|^2 \right\}^{1/2} \left\{ \int_a^b |g - h|^2 \right\}^{1/2} + \|g - h\|_2{}^2$$
$$= \|f - g\|_2{}^2 + 2\|f - g\|_2 \|g - h\|_2 + \|g - h\|_2{}^2 = \left(\|f - g\|_2 + \|g - h\|_2 \right)^2.$$

Thus, $\left(\left(\int_a^b |f-h|^2 d\alpha\right)^{1/2}\right)^2 \leq \left(\|f-g\|_2 + \|g-h\|_2\right)^2$.

Therefore, $\|f-h\| = \left(\int_a^b |f-h|^2 d\alpha\right)^{1/2} \leq \|f-g\|_2 + \|g-h\|_2$. ■

Exercise 6.14: Let

$$f(x) = \int_x^{x+1} \sin\left(e^t\right) dt$$

Claim 1: $e^x|f(x)| < 2$.

Proof: Let $f(x) = \int_x^{x+1} \sin\left(e^t\right) dt$. Show that $e^x|f(x)| < 2$. First we find $f(x)$. Substituting $y = e^t$ so that $\ln(y) = \ln\left(e^t\right) = t$, and, thus, $dt = \frac{1}{y}dy$. So, $f(x) = \int_x^{x+1} \frac{\sin(y)}{y} dy$. Now, using integration by parts, let $\frac{1}{y} = u$ and $dv = \sin(y)dy$. Then, by change of variables, $f(x) = uv - \int v du = \frac{-\cos(y)}{y}\Big|_x^{x+1} - \int_x^{x+1} \frac{\cos(y)}{y^2} dy = \frac{-\cos(e^t)}{e^t}\Big|_x^{x+1} - \int_x^{x+1} \frac{\cos(e^t)}{(e^t)^2} dt = \frac{\cos(e^x)}{e^x} - \frac{\cos(e^{x+1})}{e^{x+1}} - \int_{e^x}^{e^{x+1}} \frac{\cos(s)}{s^2} ds$, with $s \in [e^{x+1}, e^x]$. As $-1 \leq \cos(s) \leq 1$ for all $s \in (-\infty, \infty)$, $-\frac{1}{s^2} \leq \frac{\cos(s)}{s^2} \leq \frac{1}{s^2}$ for all $s \in (-\infty, \infty)$. Thus, $e^x f(x) \geq e^x \left[\frac{\cos(e^x)}{e^x} - \frac{\cos\left(e^{x+1}\right)}{e^{x+1}} - \int_{e^x}^{e^{x+1}} \frac{1}{s^2}\right] = \cos\left(e^x\right) - \frac{\cos\left(e^{x+1}\right)}{e} + \frac{1}{e} - 1 = \left(\cos\left(e^x\right) - 1\right) + \frac{\left(1-\cos\left(e^{x+1}\right)\right)}{e} > -2$. And, $e^x f(x) \leq e^x \left[\frac{\cos(e^x)}{e^x} - \frac{\cos\left(e^{x+1}\right)}{e^{x+1}} + \int_{e^x}^{e^{x+1}} \frac{1}{s^2}\right] = \cos\left(e^x\right) - \frac{\cos\left(e^{x+1}\right)}{e} - \frac{e^x}{e^{x+1}} + \frac{e^x}{e^x} = \cos\left(e^x\right) + 1 - \frac{\left(1+\cos\left(e^{x+1}\right)\right)}{e} < 2$. Hence $-2 < e^x f(x) < 2$. Thus, $|e^x f(x)| = e^x|f(x)| < 2$· ■

Claim 2: $e^x f(x) = \cos\left(e^x\right) - e^{-1}\cos\left(e^{x+1}\right) + r(x)$, where $|r(x)| < Ce^{-x}$, for some constant C.

Proof: Recall from the proof for Claim 1 that $e^x f(x) = \left[\frac{\cos(e^x)}{e^x} - \frac{\cos\left(e^{x+1}\right)}{e^{x+1}} - \int_x^{x+1} \frac{\cos(y)}{y^2} dy\right] e^x = \cos\left(e^x\right) - \frac{\cos\left(e^{x+1}\right)}{e} - e^x \int^x \frac{\cos\left(e^t\right)}{x+e^x)^x} dt$. For $t \in [x, x+1], e^t \geq e^x$, since e is an increasing function. We claim that $r(t) = -e^x \int_x^{x+1} \frac{\cos\left(e^t\right)}{(e^t)^2} dt$.

$$\text{Then, } |r(t)\,| = \left|e^x \int_x^{e^{x+1}} \frac{\cos\left(e^t\right)}{\left(e^t\right)^2} dt\right| = \left|\int_x^{x+1} e^x \frac{\cos\left(e^t\right)}{\left(e^t\right)^2 dt}\right| \leq \left|\int_x^{x+1} \frac{e^t \cos\left(e^t\right)}{\left(e^t\right)^2} dt\right|$$

$$= \left|\int_x^{x+1} \frac{\cos\left(e^t\right)}{e^t} dt\right| < \left|\int_x^{x+1} \frac{1}{e^t} dt\right| = \left|\int_x^{x+1} e^{-t} dt\right| = \left|-e^{-t}\,|_x^{x+1}\right| = |e^{-x} - e^{-(x+1)}| = |\frac{1}{e^x} - \frac{1}{e^{x+1}}|$$

$$= e^{-x}\left|1 - \frac{1}{e}\right| = e^{-x}\left(1 - \frac{1}{e}\right) = Ce^{-x}, \text{ with } C = 1 - \frac{1}{e}. \text{ Thus, the proof is complete. } ■$$